Maria Ios Glarou

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Education

Dec. 2025 – June 2026 **Valorization of Master's Program**, EPFL
Machine Learning and Optimization Laboratory (MLO), Prof. Martin Jaggi

2018 – 2025 **Integrated Master's in Electrical and Computer Engineering**, National Technical University of Athens (NTUA)
Major: Computer Science • **GPA**: 9.8/10
Equivalent to an M.Sc. and M.Eng. degree.
Thesis: *On Learning What to Learn: Adaptive Data Mixtures for Robust Multi-Target LLM Pretraining*
Supervised by Prof. Martin Jaggi (EPFL) and Prof. Petros Maragos (NTUA), during a 7-month paid research appointment at EPFL.

Research Experience

Spring semester of 2027 **ETH Zürich & MPI-IS Tübingen** Research Visitor (upcoming)

- Host: Prof. Bernhard Schölkopf

From December 2025 **EPFL** Research Assistant

- Advisor: Prof. Martin Jaggi
- Initially explored probe-guided data selection for LLM pretraining.
- *May – October* 2026*:* Showed that data usually filtered out of pretraining can serve as an explicit negative training signal, making unwanted knowledge much harder to recover through fine-tuning (DETER, first-author paper).
- *Since October* 2026*:* Researching reinforcement learning for post-training LLMs (with Giannis Daras, MIT).

Spring semester of 2025 **EPFL** Research Assistant

- Advisor: Prof. Martin Jaggi
- Worked on data mixture optimization for LLM pretraining; contributed toward LLMs whose representations better support our objectives.
- Introduced GRAPE, a minimax-formulated method that can identify superior online data mixtures for multiple tasks by dynamically reallocating sampling toward lagging tasks; improved reasoning and low-resource multilingual performance over strong baselines.

Research Experience (continued)

Summer of 2024 — **EPFL** Research Intern

- Advisor : Prof. Ola Svensson
- Explored the use of Large Language Models (LLMs) for optimizing Mixed Integer Programming (MIP) problem formulations to improve solver efficiency.
- Investigated downsizing Integer Program (IP) instances using Neural Networks by learning a smaller surrogate problem that preserves the original's objective/constraints, solving it faster, and lifting the surrogate solution back to the full instance.

Summer of 2023 — **EPFL** Research Intern

- Advisor : Prof. Antoine Bosselut
- Explored methods to enhance the quality and relevance of images generated by state-of-the-art text-to-image models (Stable Diffusion). Used techniques to expand input text prompts using relevant commonsense knowledge and a lightweight transformer-based adapter to produce enhanced text embeddings and align them with the CLIP embedding used in Stable Diffusion. Conducted shortly before later releases (e.g., DALL·E 3) popularized integrated prompt expansion.

Work Experience

Summer of 2022 — **Hellenic Drones** Machine Learning Engineer Intern

- Worked on developing a real-time system for aerial object detection and tracking by integrating YOLOv5 with DeepSORT. Achieved 87% accuracy in aerial object classification and improved tracking performance in occlusion-prone scenarios. Our model was deployed on the company's drones for aerial surveillance.

Hellenic Drones is a startup in Athens collaborating at the time with members of the Artificial Intelligence and Learning Systems (AILS) lab at NTUA.

Volunteering

Peer Review — Reviewer for ICLR 2027

Area Chair support for the ICML 2026 review process

- Assisted Alexander Hägele (PhD student, MLO, EPFL) and Prof. Martin Jaggi (Area Chair) in verifying references in submissions.

Teaching Assistant — Greek National Olympiad in Informatics Workshops

- Explained algorithmic methods and example problems to high school students and assisted them during practice sessions held at NTUA.

These workshops are organized every year by Professors Dimitris Fotakis, Nikolaos Papaspyrou, and Aris Pagourtzis (NTUA).

Volunteer — Zurich AI Safety Day 2025

Research Publications

1. M. I. Glarou and M. Jaggi, *DETER: Suppression during pretraining for tamper resistance*, Preprint, 2026.
2. S. Fan, M. I. Glarou, and M. Jaggi, "GRAPE: Optimize data mixture for group robust multi-target adaptive pretraining," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv: 2505.20380 [cs.LG]. URL: https://arxiv.org/abs/2505.20380

Skills

Languages

- **English**: Proficient in reading, writing, and speaking (Certificate of Competency in English, Michigan Language Institute).
- **French**: Basic understanding of spoken and written French; completed 3 years of French classes during high school.
- **Greek**: Native speaker.

Coding	Python, C++, SQL, LaTeX.
Databases	Experience in MySQL, PostgreSQL, SQLite.
Web Dev	Experience in HTML, CSS, JavaScript.
Communication	Collaborated on multiple group projects at NTUA, coordinating with team members to achieve project goals on time.
Personality Traits	Curious, energetic, perspective, persistent.
Hobbies	Hiking, swimming, traveling, music, books, languages.
Misc.	Academic research.